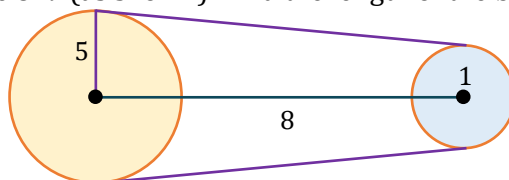




Trigonometric Ratios and Identities: RACE-01

Single Correct Question

- The radius of a circle is 30 cm. if the length of the chord of the arc is 30 cm. then the length of smaller arc of the circle is
(A) $5\pi\text{cm}$ (B) $10\pi\text{cm}$ (C) $15\pi\text{cm}$ (D) $20\pi\text{cm}$ MTR131
- A man is running around a regular hexagonal field of side length 6m, so that he always at a distance 3 metre from the nearest boundary. The length of path travelled by him in one round is –
(A) $36 + 3\pi$ (B) $36 + 2\pi$ (C) $36 + 6\pi$ (D) $18 + 3\pi$ MTR132
- A belt is tied up across two circular pulleys of radii 5m and 1m respectively whose centres are separated at a distance 8m. (as shown). Find the length of the belt required.



- (A) $\frac{22\pi}{3} + 8\sqrt{3}$ (B) $6\pi + 8\sqrt{3}$ (C) $\frac{22\pi}{3} + 4\sqrt{3}$ (D) $6\pi + 4\sqrt{3}$ MTR133
- If $\sin\theta + \operatorname{cosec}\theta = 2$, then the value of $\sin^8\theta + \operatorname{cosec}^8\theta$, is
(A) 2^8 (B) 2^4 (C) 2 (D) can't be determined MTR134
 - The value of $\operatorname{cosec}^6\alpha - \cot^6\alpha - 3\operatorname{cosec}^2\alpha \cot^2\alpha$, is
(A) 1 (B) 2 (C) 3 (D) $\frac{1}{2}$ MTR135

Multiple Correct Question

- If θ lies in first quadrant and $\cot\theta = \frac{4}{3}$, then
(A) $\sin\theta = \frac{4}{5}$ (B) $\sin\theta = \frac{3}{5}$ (C) $\cos\theta = \frac{4}{5}$ (D) $\operatorname{cosec}\theta = \frac{5}{3}$ MTR136
- Which of the following is/are correct
(A) $(\sin A + \cos A)(1 - \sin A \cos A) = \sin^3 A + \cos^3 A$.
(B) $\cos^6 A + \sin^6 A = 1 - 3 \sin^2 A \cos^2 A$.
(C) $\sqrt{\frac{1 - \sin A}{1 + \sin A}} = \sec A - \tan A$ ($-90^\circ < A < 90^\circ$)
(D) $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A} = \sec A \operatorname{cosec} A + 1$ MTR137

8. If $\sin \theta = \frac{x^2 - y^2}{x^2 + y^2}$, where $\theta \in (0, 90^\circ)$. Then -

(A) $\cos \theta = \frac{2xy}{x^2 + y^2}$ (B) $\cos \theta = \frac{2xy}{x^2 - y^2}$ (C) $\cot \theta = \frac{2xy}{x^2 + y^2}$ (D) $\cot \theta = \frac{2xy}{x^2 - y^2}$

MTR138

Subjective Type Question

9. Convert the following measurement into radians :

(a) $25^\circ 30' 30''$ (b) $10^\circ 42' 30''$ (c) $9^\circ 18' 42''$

MTR139

10. Prove that :

(i) $\sec^4 A - \sec^2 A = \tan^4 A + \tan^2 A$.

(ii) $\tan^2 A - \sin^2 A = \sin^4 A \sec^2 A = \tan^2 A \cdot \sin^2 A$

(iii) $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$

MTR140

(iv) $\frac{1}{\operatorname{cosec} A - \cot A} - \frac{1}{\sin A} = \frac{1}{\sin A} - \frac{1}{\operatorname{cosec} A + \cot A}$

(v) $\frac{\cot A \cdot \cos A}{\cot A + \cos A} = \frac{\cot A - \cos A}{\cot A \cdot \cos A}$

(vi) $\sin^8 A - \cos^8 A = (\sin^2 A - \cos^2 A)(1 - 2\sin^2 A \cos^2 A)$

MTR141

(vii) $\frac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \frac{1 + \sin A}{\cos A}$

(viii) $2 \sec^2 \alpha - \sec^4 \alpha - 2 \operatorname{cosec}^2 \alpha + \operatorname{cosec}^4 \alpha = \cot^4 \alpha - \tan^4 \alpha$.

MTR142

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	B	C	A	C	A	B,C,D	A,B,C,D	A,D

9. (a) $\left(\frac{3061}{21600}\pi\right)$; (b) $\left(\frac{257}{4320}\pi\right)$; (c) $\frac{5587}{108000}\pi$



Trigonometric Ratios and Identities: RACE-02

Single Correct Question

1. The value of $\cos 570^\circ \sin 510^\circ + \sin(-330^\circ) \cos(-390^\circ)$, is equal to

(A) 1 (B) 0 (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

MTR143

2. The value of $\tan \frac{11\pi}{3} - 2 \sin \frac{9\pi}{3} - \frac{3}{4} \operatorname{cosec}^2 \frac{\pi}{4} + 4 \cos^2 \frac{17\pi}{6}$, is equal to

(A) $\frac{3+2\sqrt{3}}{2}$ (B) $3+2\sqrt{3}$ (C) $3-2\sqrt{3}$ (D) $\frac{3-2\sqrt{3}}{2}$

MTR144

Multiple Correct Question

3. If $\sin \theta = \frac{4}{5}$ then which of the following may be correct-

(A) $\cos \theta = \frac{3}{5}$ (B) $\cos \theta = \frac{-3}{5}$ (C) $\cot \theta = \frac{-4}{3}$ (D) $\cot \theta = \frac{4}{3}$

MTR145

4. If $3 \sec^4 \theta + 8 = 10 \sec^2 \theta$, then the values of $\tan \theta$ can be-

(A) 1 (B) -1 (C) $\frac{-1}{\sqrt{3}}$ (D) $\sqrt{3}$

MTR146

5. If $\tan \theta = \frac{2x^2 + x + 1}{2x + 1}$, then:

(A) $\sin \theta = \pm \frac{2x(x+1)}{2x^2 + 2x + 1}$ (B) $\sin \theta = \pm \frac{2x+1}{2x^2 + 2x + 1}$
 (C) $\cos \theta = \pm \frac{2x(x+1)}{2x^2 + 2x + 1}$ (D) $\cos \theta = \pm \frac{2x+1}{2x^2 + 2x + 1}$

MTR147

6. If $\tan^2 \theta + \sec \theta = 5$, then value of $\cos \theta$ can be-

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

MTR148

Integer Type Question

7. If $\tan \theta = \frac{1}{\sqrt{7}}$, find the value of $4 \left(\frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta} \right)$, is

MTR149

8. If $\tan\theta + \sec\theta = 1.5$, then value of $13\sin\theta$, is where $0 < \theta < \frac{\pi}{2}$

MTR150

9. If $\cos\theta = -\frac{1}{2}$ and $\pi < \theta < \frac{3\pi}{2}$, then find the value of $4\tan^2\theta - 3\operatorname{cosec}^2\theta$.

MTR151

Subjective Type Question

10. **Evaluate:**

(a) $\sin 420^\circ \cos 390^\circ + \cos(-300^\circ) \sin(-330^\circ)$

MTR152

(b) $\tan 225^\circ \cot 405^\circ + \tan 765^\circ \cot 675^\circ$

MTR153

11. Express the following quantities in terms of the ratios of a positive angle, which is less than 45°

(a) $\sin(-65^\circ)$ (b) $\cos(-928^\circ)$ (c) $\tan 1145^\circ$ (d) $\cot(-1054^\circ)$

MTR154

12. What is the sign of $\sin A + \cos A$ for the following values of A ?

(a) 140° (b) -356° (c) -1125° .

MTR155

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	B	D	A,B	A,B,C	A,D	B,D	3	5	8

10. (a) 1; (b) 0
 11. (a) $-\cos(25^\circ)$; (b) $-\cos(28^\circ)$; (c) $\cot 25^\circ$; (d) $\cot(26^\circ)$
 12. (a) negative; (b) positive; (c) zero



Trigonometric Ratios and Identities: RACE-03

Single Correct Type Question:

1. The value of
- $\sin 105^\circ + \cos 105^\circ =$

(A) 1 (B) 0 (C) $\frac{3}{2}$ (D) $\frac{1}{\sqrt{2}}$

MTR156

2. The value of
- $\tan\left(\frac{\pi}{4} + \theta\right) \times \tan\left(\frac{3\pi}{4} + \theta\right)$
- , (wherever defined) equals

(A) 1 (B) $\frac{1}{2}$ (C) $\sqrt{3}$ (D) -1

MTR157

3. The number of solutions of the equation
- $\sin(\pi x) + x^2 + 1 = 2x$
- , is

(A) 0 (B) 1 (C) 2 (D) 3

MTR158

4. If
- $\tan \alpha = \frac{5}{6}$
- and
- $\tan \beta = \frac{1}{11}$
- , then
- $\alpha + \beta =$

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

MTR159

Multiple Correct Type Question

5. Which of the following is/are correct

(A) The equation $\sin \theta = x + \frac{1}{x}$ is impossible if x be real.

(B) The equation $\sec^2 \theta = \frac{4xy}{(x+y)^2}$ is only possible when $x = y$.

(C) The number of real solutions of the equation $\sin(e^x) = 2^x + 2^{-x}$ is 0

(D) The number of real solutions of the equation $\sin(e^x) = 2^x + 2^{-x}$ is 2

MTR160

6. If
- $\sin A = \frac{3}{5}$
- and
- $\cos B = \frac{9}{41}$
- ,
- $0 < A, B < \frac{\pi}{2}$
- , then

(A) $\sin(A + B) = \frac{187}{205}$ (B) $\sin(A - B) = \frac{-133}{205}$

(C) $\cos(A + B) = \frac{-84}{205}$ (D) $\cos(A - B) = \frac{156}{205}$

MTR161

7. If $x + y = 45^\circ$, which of the following is/are correct-

(A) $(1 + \tan x)(1 + \tan y) = 2$

(B) $(1 + \tan x)(1 + \tan y) = 1$

(C) $(\cot x - 1)(\cot y - 1) = 2$

(D) $(\cot x - 1)(\cot y - 1) = 1$

MTR162

8. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, then

(A) $\tan(2A + B) = \frac{9}{13}$

(B) $\tan(2A + B) = 3$

(C) $\tan(2A - B) = \frac{9}{13}$

(D) $\tan(2A - B) = 3$

MTR163

Subjective Type Question

9. Draw graphs of

(a) $y = \sin 2x$

(b) $y = 2\cos 3x$

(c) $y = 4 \tan x$

MTR164

10. Prove that

(a) $\sin(45^\circ + A) \cos(45^\circ - B) + \cos(45^\circ + A) \sin(45^\circ - B) = \cos(A - B)$.

(b) $\cos(\alpha + \beta)\cos\gamma - \cos(\beta + \gamma)\cos\alpha = \sin\beta \sin(\gamma - \alpha)$

(c) $\sin(n + 1)A \sin(n - 1)A + \cos(n + 1)A \cos(n - 1)A = \cos 2A$

MTR165

(d) $1 + \tan A \tan \frac{A}{2} = \tan A \cot \frac{A}{2} - 1 = \sec A$

(e) $\tan 3A \tan 2A \tan A = \tan 3A - \tan 2A - \tan A$.

MTR166

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	D	D	C	B	A,B,C	A,B,C,D	A,C	B,C



Trigonometric Ratios and Identities: RACE-04

Subjective Type Question

1. Simplify $\frac{\sin 75^\circ - \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ}$ MTR167
2. Simplify $2 \sin 54^\circ \sin 66^\circ$ MTR168
3. $\cos(36^\circ - A) \cos(36^\circ + A) + \cos(54^\circ + A) \cos(54^\circ - A) = \cos 2A.$ MTR169
4. $\sin(45^\circ + A) \sin(45^\circ - A) = \frac{1}{2} \cos 2A$ MTR170
5. $\sin(\beta - \gamma) \cos(\alpha - \delta) + \sin(\gamma - \alpha) \cos(\beta - \delta) + \sin(\alpha - \beta) \cos(\gamma - \delta) = 0.$ MTR171
6. $2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0$ MTR172
7. $\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta} = \tan 2\theta$ MTR173
8. $\frac{\sin 2A + \sin 2B}{\sin 2A - \sin 2B} = \frac{\tan(A+B)}{\tan(A-B)}$ MTR174
9. $\cos(A+B) + \sin(A-B) = 2 \sin(45^\circ + A) \cos(45^\circ + B)$ MTR175
10. $\frac{\cos 3A - \cos A}{\sin 3A - \sin A} + \frac{\cos 2A - \cos 4A}{\sin 4A - \sin 2A} = \frac{\sin A}{\cos 2A \cos 3A}$ MTR176
11. $\frac{\cos 3\theta + 2 \cos 5\theta + \cos 7\theta}{\cos \theta + 2 \cos 3\theta + \cos 5\theta} = \cos 2\theta - \sin 2\theta \tan 3\theta$ MTR177
12. $\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A$ MTR178
13. $\frac{\sin(A-C) + 2 \sin A + \sin(A+C)}{\sin(B-C) + 2 \sin B + \sin(B+C)} = \frac{\sin A}{\sin B}$ MTR179

14. $\frac{\sin A + \sin B}{\sin A - \sin B} = \tan \frac{A+B}{2} \cot \frac{A-B}{2}$ **MTR180**
15. $\frac{\sin A - \sin B}{\cos B - \cos A} = \cot \frac{A+B}{2}$ **MTR181**
16. $\frac{\cos(A+B+C) + \cos(-A+B+C) + \cos(A-B+C) + \cos(A+B-C)}{\sin(A+B+C) + \sin(-A+B+C) - \sin(A-B+C) + \sin(A+B-C)} = \cot B$ **MTR182**
17. $\cos\left\{\theta + \left(n - \frac{3}{2}\right)\phi\right\} - \cos\left\{\theta + \left(n + \frac{3}{2}\right)\phi\right\} = 2\sin\frac{3\phi}{2} \cdot \sin(\theta + n\phi)$ **MTR183**
18. $\sin\frac{\theta}{2}\sin\frac{7\theta}{2} + \sin\frac{3\theta}{2}\sin\frac{11\theta}{2} = \sin 2\theta \sin 5\theta$ **MTR184**
19. $\cos 2\theta \cos \frac{\theta}{2} - \cos 3\theta \cos \frac{9\theta}{2} = \sin 5\theta \sin \frac{5\theta}{2}$ **MTR185**
20. $\frac{2\sin(A-C)\cos C - \sin(A-2C)}{2\sin(B-C)\cos C - \sin(B-2C)} = \frac{\sin A}{\sin B}$ **MTR186**
21. $\frac{\sin A \sin 2A + \sin 3A \sin 6A + \sin 4A \sin 13A}{\sin A \cos 2A + \sin 3A \cos 6A + \sin 4A \cos 13A} = \tan 9A$ **MTR187**
22. $\frac{\cos 2A \cos 3A - \cos 2A \cos 7A + \cos A \cos 10A}{\sin 4A \sin 3A - \sin 2A \sin 5A + \sin 4A \sin 7A} = \cot 6A \cot 5A$ **MTR188**

ANSWER KEY

1. $\frac{1}{\sqrt{3}}$ 2. $\cos 12^\circ - \cos 120^\circ$



Trigonometric Ratios and Identities: RACE-05

Subjective Type Question

Prove that :

1. $\frac{\sin 2A}{1 + \cos 2A} = \tan A$ (Remember) MTR189
2. $\frac{\sin 2A}{1 - \cos 2A} = \cot A$ (Remember) MTR190
3. $\tan A - \cot A = -2 \cot 2A$. (Remember) MTR191
4. $\frac{1 + \sin 2\theta + \cos 2\theta}{1 + \sin 2\theta - \cos 2\theta} = \cot \theta$ MTR192
5. $\tan A + \cot A = 2 \operatorname{cosec} 2A$ MTR193
6. $\frac{1 - \cos A + \cos B - \cos(A+B)}{1 + \cos A - \cos B - \cos(A+B)} = \tan \frac{A}{2} \cot \frac{B}{2}$ MTR194
7. $\frac{\sec 8A - 1}{\sec 4A - 1} = \frac{\tan 8A}{\tan 2A}$ MTR195
8. $\cot(A + 15^\circ) - \tan(A - 15^\circ) = \frac{4 \cos 2A}{1 + 2 \sin 2A}$ MTR196
9. $\frac{\sin(n+1)A + 2 \sin nA + \sin(n-1)A}{\cos(n-1)A - \cos(n+1)A} = \cot \frac{A}{2}$ MTR197
10. $\cos^3 2\theta + 3 \cos 2\theta = 4(\cos^6 \theta - \sin^6 \theta)$ MTR198
11. $\sec^2 A(1 + \sec 2A) = 2 \sec 2A$ MTR199
12. $\operatorname{cosec} A - 2 \cot 2A \cos A = 2 \sin A$ MTR200
13. $\frac{2 \cos 2^n \theta + 1}{2 \cos \theta + 1} = (2 \cos \theta - 1)(2 \cos 2\theta - 1)(2 \cos 2^2 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1)$ MTR201



Trigonometric Ratios and Identities: RACE-06

Single Correct Question

1. If $\cos x + \sin x = a$, $\left(-\frac{\pi}{2} < x < -\frac{\pi}{4}\right)$, then $\cos 2x$ is equal to

(A) a^2 (B) $a\sqrt{2-a}$ (C) $a\sqrt{2+a}$ (D) $a\sqrt{2-a^2}$

MTR202

Subjective Type Question

2. If $\sin \theta = \frac{1}{2}$ and $\sin \phi = \frac{1}{3}$, find the values of $\sin(\theta + \phi)$ and $\sin(2\theta + 2\phi)$.

MTR203

3. If $\cos \alpha = \frac{11}{61}$ and $\sin \beta = \frac{4}{5}$, find the values of $\sin^2 \frac{\alpha - \beta}{2}$ and $\cos^2 \frac{\alpha + \beta}{2}$, the angles α and β being positive acute angles.

MTR204

4. Find the values of

(a) $\sin 7\frac{1}{2}^\circ$ (b) $\cos 7\frac{1}{2}^\circ$ (c) $\tan 11\frac{1}{4}^\circ$

MTR205

5. If $\sin \theta + \sin \phi = a$ and $\cos \theta + \cos \phi = b$, find the value of $\tan \frac{\theta - \phi}{2}$.

MTR206

Prove that :

6. (a) $\cot \theta \cot(60^\circ - \theta) \cot(60^\circ + \theta) = \cot 3\theta$

MTR207

(b) $\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$

MTR208

(c) $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

MTR209

7. (a) $\sin \alpha \sin(60^\circ - \alpha) \sin(60^\circ + \alpha) = \frac{1}{4} \sin 3\alpha$

MTR210

(b) $\cos \alpha \cos(60^\circ - \alpha) \cos(60^\circ + \alpha) = \frac{1}{4} \cos 3\alpha$

MTR211

(c) $\cot \alpha + \cot(60^\circ + \alpha) - \cot(60^\circ - \alpha) = 3\cot 3\alpha$.

MTR212

Trigonometric Ratios and Identities

8. $\cos 6\alpha = 32\cos^6\alpha - 48\cos^4\alpha + 18\cos^2\alpha - 1.$

MTR213

9. $(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 = 4\sin^2\frac{\alpha - \beta}{2}$

MTR214

10. $\tan\left(45^\circ + \frac{A}{2}\right) = \sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sec A + \tan A$

MTR215

11. $\cos^2\alpha + \cos^2(\alpha + 120^\circ) + \cos^2(\alpha - 120^\circ) = \frac{3}{2}$

MTR216

12. $\sin^4\frac{\pi}{8} + \sin^4\frac{3\pi}{8} + \sin^4\frac{5\pi}{8} + \sin^4\frac{7\pi}{8} = \frac{3}{2}$

MTR217

ANSWER KEY

1. (D) 2. $\frac{\pm 2\sqrt{2} \pm \sqrt{3}}{6}; \frac{\pm 7\sqrt{3} \pm 4\sqrt{2}}{18}$
3. $\frac{16}{305}; \frac{49}{305}$ 4. (a) $\frac{\sqrt{4 - \sqrt{2} - \sqrt{6}}}{2\sqrt{2}}$ (b) $\frac{\sqrt{4 + \sqrt{2} + \sqrt{6}}}{2\sqrt{2}}$ (c) $\sqrt{4 + 2\sqrt{2}} - (\sqrt{2} + 1)$
5. $\pm \sqrt{\frac{4 - a^2 - b^2}{a^2 + b^2}}$



Trigonometric Ratios and Identities: RACE-07

Subjective Type Question

1. Find the value of

(a) $\sin \frac{\pi}{10} + \sin \frac{13\pi}{10}$

(b) $\cos^2 48^\circ - \sin^2 12^\circ$

MTR218

2. $\sin \frac{\pi}{5} \sin \frac{2\pi}{5} \sin \frac{3\pi}{5} \sin \frac{4\pi}{5}$

MTR219

3. $\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ$

MTR220

4. $\cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{3\pi}{15} \cos \frac{4\pi}{15} \cos \frac{5\pi}{15} \cos \frac{6\pi}{15} \cos \frac{7\pi}{15}$

MTR221

5. If $A = \cos \frac{\pi}{9} \cos \frac{4\pi}{9} \cos \frac{16\pi}{9}$, $B = \cos \frac{2\pi}{9} \cos \frac{8\pi}{9} \cos \frac{32\pi}{9}$, then value of $\left| \frac{1}{16AB} \right|$ is

MTR222

6. In any circle prove that the chord which subtends 108° at the centre is equal to the sum of the two chords which subtend angles of 36° and 60° .

MTR223

7. If $x + y + z = xyz$ prove that

(a) $\frac{3x-x^3}{1-3x^2} + \frac{3y-y^3}{1-3y^2} + \frac{3z-z^3}{1-3z^2} = \frac{3x-x^3}{1-3x^2} \cdot \frac{3y-y^3}{1-3y^2} \cdot \frac{3z-z^3}{1-3z^2}$

MTR224

(b) $x(1-y^2)(1-z^2) + y(1-z^2)(1-x^2) + z(1-x^2)(1-y^2) = 4xyz$.

MTR225

For Question Number 8 to 15, consider $A + B + C = \pi$. Prove that

8. $\sin 2A + \sin 2B - \sin 2C = 4 \cos A \cos B \sin C$

MTR226

9. $\cos 2A + \cos 2B - \cos 2C = 1 - 4 \sin A \sin B \cos C$

MTR227

10. $\sin^2 A + \sin^2 B - \sin^2 C = 2 \sin A \sin B \cos C$

MTR228

11. $\cos^2 A + \cos^2 B - \cos^2 C = 1 - 2 \sin A \sin B \cos C$

MTR229

$$12. \quad \sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} - \sin^2 \frac{C}{2} = 1 - 2 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$$

MTR230

$$13. \quad \sin(B + 2C) + \sin(C + 2A) + \sin(A + 2B) = 4 \sin \frac{B-C}{2} \sin \frac{C-A}{2} \sin \frac{A-B}{2}.$$

MTR231

$$14. \quad \frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

MTR232

$$15. \quad \sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C) = 4 \sin A \sin B \sin C$$

MTR233

ANSWER KEY

- | | | | | | | | | |
|----|--------------------|----------------------------|----|----------------|----|-----|----|-----------------|
| 1. | (a) $-\frac{1}{2}$ | (b) $\frac{\sqrt{5}+1}{8}$ | 2. | $\frac{5}{16}$ | 3. | (1) | 4. | $\frac{1}{2^7}$ |
| 5. | (4) | | | | | | | |



Trigonometric Ratios and Identities: RACE-08

Single Correct Question

1. If $S = \cos^2 \frac{\pi}{n} + \cos^2 \frac{2\pi}{n} + \dots + \cos^2 \frac{(n-1)\pi}{n}$, then S equals (where $n \geq 2, n \in N$)
- (A) $\frac{n}{2(n+1)}$ (B) $\frac{1}{2(n-1)}$ (C) $\frac{n-2}{2}$ (D) $\frac{n}{2}$

MTR234

Integer Type Question

2. Find the minimum value of $4\sec^2 x + 9\operatorname{cosec}^2 x$
3. Evaluate $\sin \frac{\pi}{n} + \sin \frac{3\pi}{n} + \sin \frac{5\pi}{n} + \dots$ to n terms

MTR235

MTR236

Subjective Type Question

4. Find minimum value of
(i) $3\sin^2 x + 27\operatorname{cosec}^2 x$ (ii) $27\sin^2 x + 3\operatorname{cosec}^2 x$
5. Find maximum and minimum value of $5\cos\theta + 3\sin\left(\theta + \frac{\pi}{6}\right)$ for all real values of θ .
6. Find the minimum value of $\cos\theta + \cos 2\theta$ for all real values of θ .
7. Find maximum and minimum value of $\cos^2 \theta - 6\sin\theta\cos\theta + 3\sin^2 \theta + 2$.
8. If $\alpha + \beta = 90^\circ$, then find the maximum value of $\sin\alpha \cdot \sin\beta$.
9. Find sum of the following series :
(a) $\cos \frac{\pi}{2n+1} + \cos \frac{3\pi}{2n+1} + \cos \frac{5\pi}{2n+1} + \dots$ up to n terms.
(b) $\sin 2\alpha + \sin 3\alpha + \sin 4\alpha + \dots + \sin n\alpha$, where $(n+2)\alpha = 2\pi$
10. Prove that : $\cos \frac{\pi}{11} + \cos \frac{3\pi}{11} + \cos \frac{5\pi}{11} + \cos \frac{7\pi}{11} + \cos \frac{9\pi}{11} = \frac{1}{2}$

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ANSWER KEY

1. (C) 2. (25) 3. (0) 4. (i) 30 (ii) 18
5. (7 & -7) 6. $-\frac{9}{8}$ 7. $4 + \sqrt{10}, 4 - \sqrt{10}$ 8. $\frac{1}{2}$
9. (a) $\frac{1}{2}$ (b) (0)